Principles of Machine Learning

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Fall, 2026

Exercise Session 3: Learning theory and practical aspects in ML

3.1 The RPNI Algorithm

Let $\{aa, ab, ba\}$ be the positive examples and $\{baa\}$ the negative examples of strings over the alphabet $\Sigma = \{a, b\}$. Apply the RPNI algorithm to find a DFA that correctly labels these instances. Which regular expression does this correspond to?

3.2 VC-Dimension

Consider the instance space $\mathbb{R}^2$ and take as a hypothesis space the set of all hypotheses of the form "everything inside rectangle R is positive and everything outside it is negative", where R can be any axis-parallel rectangle. This means that the sides of the rectangle are parallel to the X- and Y-axis. Show that the VC-dimension of this hypothesis space is at least 3. Optionally show that the VC-dimension of this hypothesis space is at least 4.

Now consider a similar hypothesis space, where for each hypothesis that states "everything inside rectangle R is positive and everything outside it is negative" there is also a hypothesis "everything outside R is positive and everything inside is negative". Show that the VC-dimension of this hypothesis space is at least 4.

3.3 The Pizza Problem

Suppose pizzas can be composed of the following 8 ingredients: *cheese, ham, tomatoes, mushrooms, peppers, salami, olives* and *shrimp*. Further assume that the concept "pizzas that Wayne likes" can be expressed as a conjunction of "contains I" literals, with I a single ingredient. For instance, "Wayne likes all pizzas that contain peppers and salami" is a possible concept, whereas "Wayne likes all pizzas that contain peppers and no olives" is not. Show that the VC-dimension of this hypothesis space is at least 3 and at most 8.

3.4 Sample Complexity

Using the concept of VC-dimension, compute a lower bound for the number of examples that may be needed to train a 2-input perceptron such that with 90% certainty it learns a hypothesis with true error smaller than 5%.

Hint: The first step is to find a lower bound for the VC-dimension of a perceptron. Then, look in the slides for a formula linking VC-dimension to number of required samples.
Hint 2: A 2-input perceptron is equivalent to a 2D linear separator.

3.5 Feature Selection

Assume we want to classify instances into three classes c_1, c_2 and c_3. Our original input has three features a_1, a_2 and a_3 of which we want to keep only two. Feature a_1 is similar for instances of classes c_1 and c_2, but different for c_3 instances. Feature a_2 is similar for instances of classes c_2 and c_3, but different for c_1 instances. Instances of all three classes have similar a_3. Let f_i be the fraction of instances of class c_i in the training dataset. We know that $f_1 > f_2 > f_3$.

When performing greedy forward feature selection, which feature will get selected first? Which second? Why?

3.6 Fatigue Prediction

Take a look at this paper[1] which can be found at `https://dl.acm.org/citation.cfm?id=3219864`. The aim of this research is to predict using machine learning techniques how fatigued runners are after performing with the aim of preventing injuries.

For this purpose, five accelerometers were attached to the runners, one at the back, one at each wrist and one at each leg. The data of all these sensors can be combined to predict the runner's fatigue. It is also possible to fuse this data together before using it to predict. Furthermore some general statistical features and some more advanced sport science features are calculated on these temporal data.

Read the paper and try to think about the following questions:

1. Which sensor locations are most useful?
2. Does fusing data affect prediction and run time performance?
3. Are the advanced sport science features useful?

3.7 Feature Selection

During the lectures, we have discussed multiple strategies for doing feature selection. In this practical exercise, you'll explore the difference between three strategies: Principal Component Analysis, Single Variable Classifier-based Feature Selection and Backward Feature Selection.

Download the 'feature_selection_assignment' notebook and complete the gaps. Why do you think one feature selection method performed better than the others? How does the nature of the dataset influence the effectiveness of each method?

3.8 Comparing Models

Every ML model has its own strengths and weaknesses. Often, it is not obvious which model would work best on a certain dataset. Hence, it is important to be able to compare models. In this exercises, you'll perform a paired t-test to compare two machine learning models using 10-fold cross validation. Download the 't_test_assignment' notebook and fill in the gaps.

Answer the following questions:

- Based on the p-value, can we reject the null hypothesis that both models have the same performance? Discuss what a p-value lower than 0.05 would mean in this context.
- Why is a paired t-test used in this case, rather than an independent t-test?
- How does cross-validation help ensure the reliability of the comparison between the two models?

[1] Fatigue Prediction in Outdoor Runners Via Machine Learning and Sensor Fusion; Tim Op De Beéck, Wannes Meert, Kurt Schütte, Benedicte Vanwanseele, Jesse Davis; 2018